

Albert Cao

✉ albertcao@college.harvard.edu ◦ ☎ (301) 356-1522 ◦ 🌐 abeot ◦ in abeot ◦ 🌐 abeot.github.io

Education and Coursework

Harvard University

Aug 2026 – May 2030 (expected)

Intended Joint Concentration in Mathematics and Computer Science.

Experience

Intern at Axilon (Startup)

January 2026 – present

Built an agentic RAG system that generates Ignition SCADA projects from natural-language prompts, replacing manual engineering workflows.

Research Intern at University of Michigan

March 2023 – January 2026

Developed deep learning models based for drug toxicity prediction, synthesizing biological and molecular features with previously neglected factors in drug-target interaction.

Intern at Johns Hopkins Applied Physics Laboratory

June 2024 – May 2025

Developed SMT-based control-flow graph recovery software and LG-PEACE, a low-level vulnerability analysis framework.

Publications

Jusko, N. M., Karekar, N., **Cao, A.**, & Sun, D. (2026). AI and machine learning for predicting in vitro/preclinical drug toxicities and adverse events in clinical patients. Acta Advance online publication. <https://doi.org/10.1016/j.apsb.2026.05.028>

Jusko, N.M., **Cao, A.** and Sun, D. (2026), Machine Learning Model Predicts Clinical Adverse Events of Small Molecule Kinase Inhibitors in Cancer Patients Using On-/Off-Target Engagement and Tissue Selectivity. Clin Pharmacol Ther. <https://doi.org/10.1002/cpt.70343>

Cao, A., Zhang, L., Bu, Y., & Sun, D. (2024), Machine Learning Prediction of On/Off Target-driven Clinical Adverse Events. Pharm Res. 2024 Aug;41(8):1649-1658. <https://doi.org/10.1007/s11095-024-03742-x>

Academic Competition Awards

USA Math Olympiad Silver Medalist

2026

Placed among the top 40 students on the USA Mathematical Olympiad, the nation's premier math olympiad.

Regeneron Science Talent Search Scholar

2026

Placed among the top 300 scholars in the 85th Regeneron Science Talent Search.

AMC, AIME and USAMO Participant

2025

Scored a 150/150 on the AMC 12, 12/15 on the AIME, and 21/42 on USAMO.

USA Junior Math Olympiad Honors

2024

Placed among the top 50 students on USA Junior Mathematical Olympiad, the nation's premier math olympiad.

USA Computing Olympiad, Gold Division

2023

Extracurriculars

Division I Fencer

DC Fencers Club, 2017 – present

A-rated fencer, top 50 nationally in Cadet Men's Epee, top 100 nationally in Junior Men's Epee.

Skills

Languages Python, Java, C++, Julia, TypeScript, C, Assembly

ML / AI PyTorch, scikit-learn, agentic pipelines, RAG

Web React, Electron, REST APIs

Security Ghidra, SMT solvers, control-flow analysis

Tools Lean 4, $\LaTeX$, Git